AcademyCloudFoundation...
PDF

Make notes Very detailed (full sentences, no short phrases)
✅ Including all important theoretical points
✅ Including procedural steps wherever steps are given
✅ Formatted cleanly for exam
make sure to include each and every imp point.

Below are **very detailed, exam-ready notes** from **AWS Academy Cloud Foundations – Module 7: Storage**, written in **full sentences**, with **all key theoretical concepts, features, comparisons, pricing factors, and procedural steps** clearly explained and neatly formatted.

All content is taken directly and comprehensively from the provided module material

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📘 MODULE 7 – AWS STORAGE (DETAILED EXAM NOTES)

🔹 INTRODUCTION TO AWS STORAGE

AWS provides multiple storage services designed for different types of data and access needs.

The main storage services discussed in this module are:

1. Amazon Elastic Block Store (EBS) – Block storage
2. Amazon Simple Storage Service (S3) – Object storage
3. Amazon Elastic File System (EFS) – File storage
4. Amazon S3 Glacier – Archival storage

Each service differs in performance, cost, scalability, and usage scenarios.

🧱 SECTION 1: AMAZON ELASTIC BLOCK STORE (EBS)

✅ What is Amazon EBS?

Amazon Elastic Block Store (EBS) is a block-level storage service designed to be attached to Amazon EC2 instances.

It allows users to create storage volumes that behave like hard drives.

📌 Key Characteristics of EBS

Amazon EBS:

• Provides block-level storage
• Is automatically replicated within the same Availability Zone
• Persists independently of EC2 instances
• Can be backed up using snapshots stored in Amazon S3

📍 Major Uses of EBS

Amazon EBS is commonly used for:

• Boot volumes for EC2 instances
• File system storage
• Database storage
• Enterprise applications requiring low latency

💽 Amazon EBS Volume Types

1. Solid State Drive (SSD) Volumes

a) General Purpose SSD

Used for most workloads requiring balanced performance and cost.

b) Provisioned IOPS SSD

Used for critical applications that require very high IOPS and throughput such as large databases.

2. Hard Disk Drive (HDD) Volumes

a) Throughput-Optimized HDD

Used for streaming workloads and big data applications.

b) Cold HDD

Used for infrequently accessed data where low cost is most important.

🧠 Use Case Summary

Volume Type	Best For
General Purpose SSD	Boot volumes, general apps
Provisioned IOPS SSD	Large databases
Throughput HDD	Big data, logs
Cold HDD	Archival, low-cost storage

⚙️ EBS Features

📸 Snapshots

• Point-in-time backups of volumes

• Can recreate new volumes anytime

• Stored in Amazon S3

🔐 Encryption

• EBS volumes can be encrypted

• No additional cost

• Data protected at rest and in transit

📈 Elasticity

• Volume size can be increased

• Volume type can be changed without downtime

💰 EBS Pricing Components

1. Volume Storage

You are charged per GB per month for provisioned storage.

2. IOPS Charges

• General Purpose SSD → Charged by GB

• Magnetic → Charged by number of requests

• Provisioned IOPS SSD → Charged by IOPS provisioned

3. Snapshots

Charged per GB-month stored in Amazon S3.

4. Data Transfer

• Incoming data is free

• Outgoing data across Regions is charged

📋 EBS Lab Procedure (Basic Workflow)

1. Create an EBS volume
2. Attach it to an EC2 instance
3. Configure it as a virtual disk
4. Create a snapshot
5. Restore from snapshot

✅ EBS Key Takeaways

• Provides persistent block storage

• Supports SSD and HDD

• Replicated within Availability Zone

• Supports snapshots and encryption

• Highly elastic and scalable

📦 SECTION 2: AMAZON SIMPLE STORAGE SERVICE (S3)

✅ What is Amazon S3?

Amazon S3 is an object-based cloud storage service where data is stored as objects inside buckets.

📌 Core Characteristics

Amazon S3:

• Stores data as objects in buckets
• Offers virtually unlimited storage
• Supports single objects up to 5 TB
• Designed for 99.999999999% (11 nines) durability
• Provides granular access control

📚 AMAZON S3 STORAGE CLASSES

1. S3 Standard

For frequently accessed data with high availability and performance.

Used for:
• Websites
• Mobile apps
• Content distribution
• Big data analytics

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2. S3 Intelligent-Tiering

Automatically moves data between frequent and infrequent tiers based on access patterns.

Key points:
• No performance impact
• No retrieval fees

• Monitors data access automatically
• Best for unpredictable workloads

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3. S3 Standard-Infrequent Access (IA)

For data accessed less often but needing fast retrieval.

Used for:
• Backups
• Disaster recovery

Lower storage cost but retrieval charges apply.

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4. S3 One Zone-IA

• Stores data in a single Availability Zone
• Lower cost than Standard-IA
• Used for secondary backups or recreatable data

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5. S3 Glacier

Low-cost archival storage.

Features:
• Very cheap long-term storage
• Retrieval takes minutes to hours
• Supports lifecycle policies

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6. S3 Glacier Deep Archive

• Cheapest S3 storage class
• Used for long-term regulatory and compliance data
• Retrieval within 12 hours
• Designed for multi-year retention

🌍 How S3 Stores Data

• Objects are stored redundantly across multiple facilities within a Region

• Designed for automatic scaling

• Accessible via Console, CLI, SDK, and URLs

📂 Common S3 Use Cases

• Backup and disaster recovery

• Static website hosting

• Media storage

• Software delivery

• Big data staging

💰 S3 Pricing Factors

1. Storage class type
2. Amount of stored data
3. Number of requests (GET, PUT, COPY, LIST)
4. Data transferred out of AWS

Inbound transfers are free.

✅ S3 Key Takeaways

• Fully managed object storage

• Unlimited scalability

• Pay-as-you-go pricing

• Global accessibility

• Strong security controls

📁 SECTION 3: AMAZON ELASTIC FILE SYSTEM (EFS)

✅ What is Amazon EFS?

Amazon EFS is a managed file storage system that provides shared access to files over a network.

📌 Key Features

Amazon EFS:

• Provides network-based file storage
• Supports petabyte-scale data
• Offers low-latency access
• Automatically scales
• Supports NFS v4.0 and v4.1
• Works with Linux EC2 instances

🏗️ EFS Architecture

• EFS is deployed within a VPC
• Mount targets are created in each Availability Zone
• EC2 instances connect through network interfaces
• Provides high availability and redundancy

🪜 EFS Implementation Steps

1. Create EC2 instances
2. Create EFS file system
3. Create mount targets in subnets
4. Connect EC2 to mount targets
5. Verify security and resources

📦 EFS Resources

• File system
• Mount targets
• Subnets

- Security groups
- Tags

✅ EFS Key Takeaways

- Shared file storage service
- Ideal for analytics and content systems
- Fully managed
- Scales automatically
- Pay only for used storage

🧊 SECTION 4: AMAZON S3 GLACIER

✅ What is Amazon S3 Glacier?

Amazon S3 Glacier is a very low-cost archival storage service for long-term data retention.

📌 Key Features

- Designed for archiving and backup
- 11 nines durability
- Data encrypted by default
- Supports compliance using Vault Lock
- Extremely low cost

⏱️ Retrieval Options

Type	Time
Expedited	1–5 minutes
Standard	3–5 hours
Bulk	5–12 hours

🔁 Lifecycle Policies

Amazon S3 lifecycle policies allow automatic movement of data such as:

S3 Standard → S3 IA → S3 Glacier → Deletion after set time

This helps reduce storage costs.

📊 S3 vs S3 Glacier Comparison

Feature	S3	Glacier
Access speed	Milliseconds	Minutes–hours
Cost	Higher	Very low
Usage	Active data	Archived data
Retrieval cost	Low	Higher

🔐 Glacier Security

• Encrypted using AES-256

• Access controlled via IAM

• AWS manages encryption keys

✅ Glacier Key Takeaways

• Best for long-term archival

• Lowest storage cost

• Highly secure

• Designed for compliance

• Very high durability

📌 FINAL COMPARISON SUMMARY

Service	Storage Type	Best For
EBS	Block	EC2 boot & databases
S3	Object	Web data, backups
EFS	File	Shared file systems
Glacier	Archive	Long-term storage

If you'd like, I can next:

✅ Convert this into **short revision notes**
✅ Make **important exam questions + answers**
✅ Create **comparison tables for quick memory**
✅ Provide **MCQs with explanations**

Just tell me 👍

Sources

convert this in a manner which is easily memorizable, use knemonics only where necessary,

Great — I've converted the detailed notes into a **high-retention, exam-friendly memory format** using:

✔ Chunking (small logical blocks)
✔ Simple flow patterns
✔ Mnemonics only where they actually help (not everywhere)
✔ Cause → use → feature style (how exam questions are framed)

This is designed so you can **recall in the exam in seconds**, not reread long theory.

🧠 AWS STORAGE — SUPER MEMORIZABLE NOTES

(Think: **B O F A** → Block, Object, File, Archive)

👉 **EBS = Block**

👉 **S3 = Object**

👉 **EFS = File**

👉 **Glacier = Archive**

🧱 AMAZON EBS — "BEDS" MEMORY TRICK

👉 EBS = **B E D S**

B → Block storage

E → Encrypted

D → Durable (replicated in AZ)

S → Snapshots

📌 EBS CORE IDEA

✔ Attached to EC2

✔ Works like hard disk

✔ Data persists after EC2 stops

✔ Used for OS + databases

💽 EBS VOLUME TYPES — "SS HOT & HDD COLD"

SSD = FAST workloads

• General Purpose SSD → everyday use

• Provisioned IOPS SSD → heavy databases

HDD = CHEAP & LARGE

• Throughput HDD → big data
• Cold HDD → rarely used

⚙️ EBS FEATURES — “SEES”

S → Snapshots
E → Encryption
E → Elastic size/type change
S → Safe replication in AZ

💰 EBS COST REMEMBER

You pay for:

✔ Storage size
✔ IOPS (if provisioned)
✔ Snapshots
✔ Data out

📥 Data in = FREE

📦 AMAZON S3 — “BOSS”

👉 S3 = **B O S S**

B → Buckets
O → Objects
S → Scalable unlimited
S → Super durable (11 nines)

📌 S3 CORE IDEA

✔ Object storage
✔ Internet accessible

✔ Huge scale
✔ Used for websites, backups, media

📚 S3 STORAGE CLASSES — "FAST → SMART → CHEAP → FREEZE"

Think in cost order 👇

⚡ FAST

S3 Standard → frequent data

🧠 SMART

S3 Intelligent-Tiering → auto moves data

💸 CHEAP

S3 Standard-IA → less access
S3 One Zone-IA → cheaper, one AZ

🧊 FREEZE

S3 Glacier → archive
S3 Deep Archive → long-term archive

🎯 SIMPLE MEMORY LINE:

Standard → Intelligent → IA → One Zone → Glacier → Deep Glacier

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💰 S3 COST BASED ON — "S R D"

S → Storage amount
R → Requests
D → Data out

📥 Data in = FREE

📁 AMAZON EFS — "FILES"

👉 EFS = **F I L E S**

F → File storage

I → Internet/network access

L → Linux supported

E → Elastic scaling

S → Shared between EC2s

📌 EFS CORE IDEA

✔ Multiple EC2 share same files

✔ Works like network drive

✔ Auto grows and shrinks

✔ Used for analytics, content, home folders

🪜 EFS SETUP STEPS — "C C M C V"

1. **C**reate EC2
2. **C**reate EFS
3. **M**ount targets
4. **C**onnect EC2
5. **V**erify security

🧊 AMAZON GLACIER — "ARCHIVE"

👉 Glacier = **A R C H I V E**

A → Archival storage

R → Retrieval slow

C → Cheapest

H → Highly durable

I → IAM security

V → Vault lock compliance

E → Encrypted

⏱️ GLACIER RETRIEVAL — "E S B"

E → Expedited (minutes)

S → Standard (hours)

B → Bulk (many hours)

🔁 LIFECYCLE FLOW (VERY EXAM IMPORTANT)

📦 S3 Standard

⬇

📦 S3 IA

⬇

🧊 Glacier

⬇

🗑️ Delete

(Old data moves to cheaper storage automatically)

📊 FINAL ONE-LOOK COMPARISON (SUPER MEMORIZABLE)

Service	Think	Best Use
EBS	Hard disk	OS + DB
S3	Cloud folder	Web + backup
EFS	Shared drive	Multiple EC2
Glacier	Cold storage	Archive

🎯 ONE-LINE EXAM RECALL FORMULA

👉 EBS = Disk for EC2

👉 S3 = Unlimited object storage

👉 EFS = Shared file system

👉 Glacier = Cheapest archive

If you want, I can now:

✅ Turn this into **flash cards**

✅ Make **1-page quick revision sheet**

✅ Create **likely exam MCQs**

✅ Provide **last-minute memory tricks**

Just say:

"Make quick revision sheet" 📄

or

"Give MCQs" 📝